OOO-T-00671b(GSA-FSS)
December 11, 1967
INTERIM REVISION OF
Fed. Spec. OOO-T-671a
August 18, 1955

INTERIM FEDERAL SPECIFICATION

TROWELS

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification OOO-T-671a, dated August 18, 1955.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers trowels used in spreading and smoothing materials such as plaster, cement, and asbestos or mastic paste while in a soft state.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Trowels shall be of the following types and classes as specified (see 6.1):

Type I. - Plastering.
Type II. - Cement.
  Class 1. - Midget.
  Class 2. - Regular.
Type III. - Brick.
Type IV. - Pointer.
Type V. - Mastic (notched).
  Class 1. - Linoleum.
  Class 2. - Asphalt tile.
Type VI. - Gauging.
Type VII. - Pipe covering.
Type VIII. - Margin.
Type IX. - Tuck Pointer, joint filler, or caulking.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:
  Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
  Fed. Test Method Std. No. 151 - Metals;Test Methods.

Federal Specifications:
  L-P-395 - Plastic Molding Material, Polypropylene, Injection and Extrusion.
  QQ-A-367 - Aluminum Alloy Forgings.

FSC 5120

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Military Specification:
MIL-H-15424 - Hand Tools, Packaging of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of trowels otherwise in accordance with this specification.

3.2 Design. Trowels shall consist essentially of a blade and handle, except that types I, II, and V trowels shall be provided with a mounting.

3.3 Materials. Materials used in construction of the trowels shall be as specified hereinafter. All forgings shall be of uniform quality and free from all defects and imperfections that might affect serviceability or durability.

3.3.1 Blades. Blades shall be of high-carbon tool steel of not less than AISI 1095.

3.3.2 Mounting. Mounting for types I, II, and V trowels shall be of permanent mold, or forged aluminum in accordance with QQ-A-367.

3.3.3 Handles.

3.3.3.1 Wood. Wood handles shall be of clear, straight-grained hardwood such as hickory, maple, birch, persimmon, or basswood. Handles shall be smoothly finished and free from cracks or splinters.

3.3.3.2 Plastic. Plastic handles shall be of high-impact strength in accordance with L-P-394, and shall withstand the tests specified in 4.4.4.1 and 4.4.4.2.

3.3.3.3 Aluminum. Types I, II, and V trowels may have aluminum handles as an integral part of the mounting.

3.4 Construction.

3.4.1 Blades. Blades shall be smoothly and evenly ground, and when placed on a flat surface shall show no evidence of warpage. Types I, II, and V trowels shall be rectangular in shape. Types III, IV, VI, VII, VIII, and IX shall taper from heel to toe and be provided with a tang. In addition, the blade may also taper from lengthwise centerline toward edges of the blade.

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3.4.1.1 Tang. The tang for types III, IV, VI, VII, VIII, and IX shall be integral with the blade, and suitably shaped to hold the handle securely.

3.4.2 Method of securing handles. Handles, where not integral with the mounting, shall be tightly fitted on the tang, and shall not turn or loosen under service conditions. For types III, IV, VI, VII, VII, and IX with wood handles, the end of the handle where the tang enters, shall be fitted with a permanently affixed metal ferrule. Except for type III, ferrule shall be not less than 1 inch long. The ferrule for type III shall be not less than 1-1/2 inches long. Wood handles for types I, II, and V trowels shall have a hole the full length of the handle and form a tight fit on the tang of the mounting. Handle shall be secured in position by a nut which may be tightened upon the threaded portion of the tang on the mounting over a suitable washer, or by riveting the end of the tang over a suitable washer.

3.4.3 Mounting. The mountings for types I, II, and V trowels shall be shaped to reinforce the blade and be provided with a suitable tang, or an integral handle. The mountings shall be secured to the blade by steel, flathead rivets or pins, and shall not separate from the blade when subjected to the tests of 4.4.2. If rivets or pins are used which extend through holes in the blades of types I, II, and V trowels, the heads of the rivets or pins shall be ground flush with the surface of the blade and top of mounting.

3.4.3.1 Tang. The tang for mounting of types I, II, and V shall be integral with the mounting, and properly shaped to accept the handle when not an integral part of the mounting.

3.5 Blade hardness. The blades shall be properly hardened and tempered to show a hardness of not less than 50 nor more than 55 on the Rockwell C scale.

3.6 Performance. The trowels shall withstand the test requirements of table I, and show no evidence of cracks, breaks, pulling out of rivets or pins, turning of edges, or permanent set when tested as specified in 4.4.2.

TABLE I. Performance test requirements

Type	Class	Movement each side of the perpendicular	Number of flexes	Strokes of shaper per minute
		Inches		
I		2	4,000	20
II	1	1-3/4	4,000	20
	2	1-3/4	4,000	20
III		3	3,000	20
IV		1-1/2	750	20
V		1-3/4	4,000	20
VI		1-1/2	750	20
VII		3	3,000	20
VIII		1-1/2	750	20
IX		2	1,500	20

3.7 Identification marking. The blade or mounting shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.8 Type I, plastering. Type I trowel blades shall be securely fastened to the mounting by a minimum of 10 flathead rivets or steel pins. Mounting shall be as specified in 3.3.2 and 3.4.3. Trowel shall conform to table II, and be similar to figure 1.

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TABLE II. Type I, plastering

Blade						Handle			
Length		Width		Thickness		Length		Diameter	
(Min)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
10-7/8	11-3/8	4-3/8	4-7/8	0.023	0.029	4-1/2	4-7/8	1-1/2	1-3/4

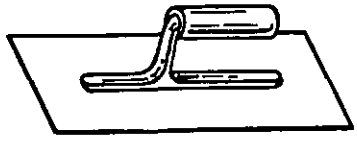


FIGURE 1. Type I, plastering.

3.9 Type II, cement.

3.9.1 Type II trowel blades shall be securely fastened to a mounting by a minimum of five flathead rivets or steel pins. Mounting shall be as specified in 3.3.2 and 3.4.3. Trowel shall conform to table III, and be similar to figure 2.

TABLE III. Type II, cement, class 1, midget trowel

Blade						Handle			
Length		Width		Thickness		Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
5-3/8	5-5/8	3-7/8	4-1/8	0.023	0.029	3-7/8	4-1/8	1-3/8	1-5/8
7-3/8	7-5/8	2-7/8	3-1/8	.023	.029	4-1/2	4-7/8	1-3/8	1-5/8

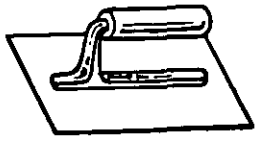


FIGURE 2. Type II, cement, class 1, midget.

3.9.2 Class 2 regular trowel. Type II, class 2 trowels shall have mountings as specified in 3.3.2 and 3.4.3. Trowel shall conform to table IV, and be similar to figure 3.

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TABLE IV. Type II, cement, class 2, regular

Blade						Handle				Number of rivets
Length		Width		Thickness		Length		Diameter		
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches	
11-7/8	12-1/8	3-7/8	4-1/8	0.023	0.029	4-1/2	4-7/8	1-1/2	1-3/4	10
13-7/8	14-1/8	3-7/8	4-1/8	.023	.029	4-1/2	4-7/8	1-1/2	1-3/4	12
15-7/8	16-1/8	3-7/8	4-1/8	.023	.029	4-1/2	4-7/8	1-1/2	1-3/4	14

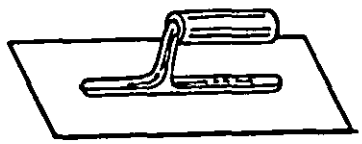


FIGURE 3. Type II, cement, class 2, regular.

3.10 Type III, brick. Type III trowels shall be of the Philadelphia, or wide heel London pattern, conform to table V, and be similar to figure 4.

TABLE V. Type III, brick.

Blade length, point to point		Blade width at widest point		Blade thickness at heel A		Handle length		Handle diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
10	10-1/2	4-7/8	5-3/8	0.065	0.085	5-1/2	6	1-1/8	1-3/8
10-1/2	11	5-1/8	5-5/8	.065	.085	5-1/2	6	1-1/8	1-3/8

A - Measurement to be taken 1/4-inch in from wingtip (see figure 4).

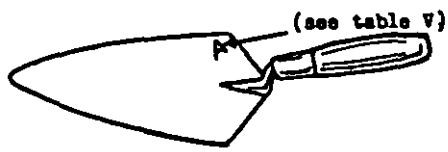


FIGURE 4. Type III, brick.

3.11 Type IV, pointer. Type IV trowels shall conform to table VI and be similar to figure 5.

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TABLE VI. Type IV, pointer

Blade length point to point		Blade width at widest point		Blade thickness at heel A		Handle length		Handle diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
4-1/2	5	2-1/4	2-1/2	0.045	0.070	4-1/4	4-5/8	3/4	1-1/8
5-1/2	6-1/4	2-3/8	2-7/8	.045	.070	4-3/8	4-7/8	1	1-1/4

A - Measurement to be taken 1/4 inch in from wingtip (see figure 5).

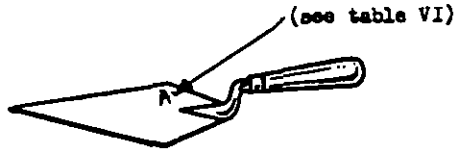


FIGURE 5. Type IV, pointer.

3.12 Type V, mastic (notched). Type V trowels shall be notched either U- or V-shaped on one or more sides and across the front end. The front end may be cut on an angle for providing even flow of mastic paste in corners and along walls. Trowel shall have a mounting as specified in 3.3.2 and 3.4.3, and be secured to the blade by not less than six flathead rivets or pins.

3.12.1 Class 1, linoleum. Class 1 trowels shall conform to table VII and be similar to figure 6.

TABLE VII. Type V, mastic, class 1 linoleum trowel

Blade						Number of notches		Handle			
Length		Width		Thickness		Front	Side	Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Min.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch			Inches	Inches	Inches	Inches
10-3/8	11-1/8	4-3/8	4-5/8	0.023	0.029	20	45	4-1/2	4-7/8	1-1/2	1-3/4



FIGURE 6. Type V, mastic, class 1, linoleum, and class 2, asphalt tile trowels.

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3.12.2 Class 2, asphalt tile. Class 2 trowels shall conform to table VIII, and be similar to figure 6.

TABLE VIII. Type V, mastic, class 2, asphalt tile trowel

Blade						Number of notches		Handle			
Length		Width		Thickness		Front	Side	Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Min.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch			Inches	Inches	Inches	Inches
10-1/4	10-5/8	4-1/2	4-5/8	0.0209	0.0269	28	70	4-1/2	4-7/8	1-1/2	1-3/4

3.13 Type VI, gauging. Type VI trowels shall conform to table IX and be similar to figure 7.

TABLE IX. Type VI, gauging

Blade						Handle			
Length		Width at widest point		Thickness at heel A		Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
6-5/8	7-1/4	3	3-3/8	0.065	0.085	4-3/4	5-1/2	7/8	1-1/4

A - Measurement to be taken 1/4 inch in from wingtip (see figure 5).

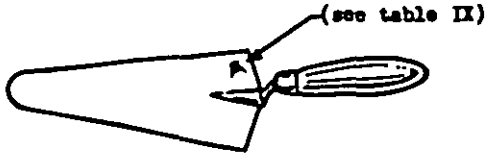


FIGURE 7. Type VI, gauging.

3.14 Type VII, pipe covering. Type VII trowels shall conform to table X and be similar to figure 8.

TABLE X. Type VII, pipe covering.

Blade						Handle			
Length		Width at widest point		Thickness at heel A		Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
9-7/8	10-1/8	1-7/8	2-1/8	0.065	0.085	4-3/4	5-1/2	1	1-1/4

A - Measurement to be taken 1/4 inch in from wingtip (see figure 8).

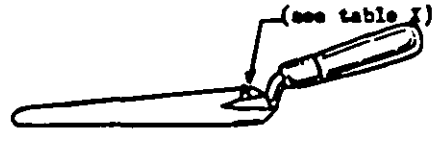


FIGURE 8. Type VII, pipe covering.

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3.15 Type VIII, margin. Type VIII trowels shall conform to table XI and be similar to figure 9.

TABLE XI. Type VIII, margin

Blade						Handle			
Length		Width at widest point		Thickness at heel A		Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inches	Inches
4-7/8	5-3/8	1-7/8	2-1/8	0.055	0.075	4-1/4	4-3/4	7/8	1-1/8
7-7/8	8-3/8	1-7/8	2-1/8	.055	.075	4-1/2	5	7/8	1-1/8

A - Measurement to be taken 1/4 inch in from wingtip (see figure 9).

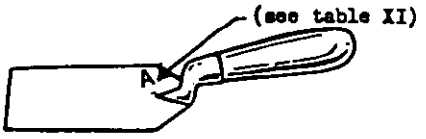


FIGURE 9. Type VIII, margin.

3.16 Type IX, tuck pointer, joint filler, or caulking. Type IX trowels shall conform to table XII and be similar to figure 10.

TABLE XII. Type IX, tuck pointer, joint filler, or caulking

Blade						Handle			
Length		Width		Thickness A		Length		Diameter	
(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)
Inches	Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inch	Inches
5-1/2	6-1/2	3/8	7/16	0.055	0.075	4-3/8	4-3/4	7/8	1-1/4
5-1/2	6-1/2	7/16	9/16	.055	.075	4-3/8	4-3/4	7/8	1-1/4
5-1/2	6-1/2	9/16	11/16	.055	.075	4-3/8	4-3/4	7/8	1-1/4
5-1/2	6-1/2	11/16	13/16	.055	.075	4-3/8	4-3/4	7/8	1-1/4
5-1/2	6-1/2	15/16	1-1/16	.055	.075	4-3/8	4-3/4	7/8	1-1/4

A - Measurement to be taken 2 inches down blade from heel (see fig. 10).

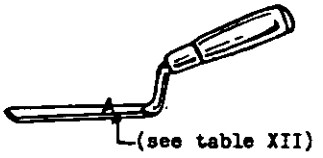


FIGURE 10. Type IX, tuck pointer, joint filler, or caulking.

3.17 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. The trowels shall be free from rust, burrs, fins, and any other defects which may impair their serviceability or durability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table XIII.

TABLE XIII. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL Expressed in terms of	Reference
Visual examination	1 each	II	2.5	Defects per hundred units	4.3.1
Dimensional examination	1 each	II	2.5	do.	4.3.2
Testing*	1 each	S-3	4.0	do.	4.4.1 thru 4.4.4
Preparation for delivery	1 container	S-2	4.0	do.	4.5

*Except as required for 4.4.2 (see 4.4).

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.4, except only one trowel from each sample unit shall be tested in accordance with 4.4.2.

4.4.1 Hardness. Testing shall be accomplished in accordance with Method 243.1 of Federal Test Method Std. No. 151.

4.4.2 Flexing test.

4.4.2.1 Type I, II, and V trowels. One of the edges of the trowels, in alignment with the handle, shall be gripped in a vise on a shaper with the edge of the trowel parallel with and set 5/16 inch below the top surface of the vise jaws. The opposite edge shall be engaged along its entire length by a fixture clamped in the tool postholder in the head of the shaper. The flexing test shall be as illustrated in figure 11, and conform to table I (see 3.6).

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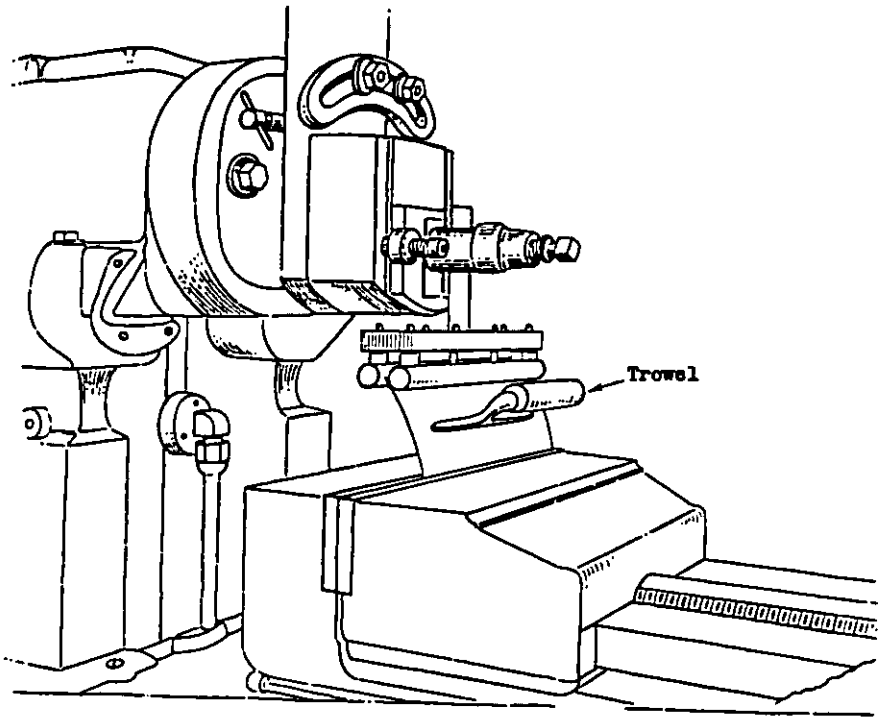


FIGURE 11.—Method of test for types I, II and V trowels.

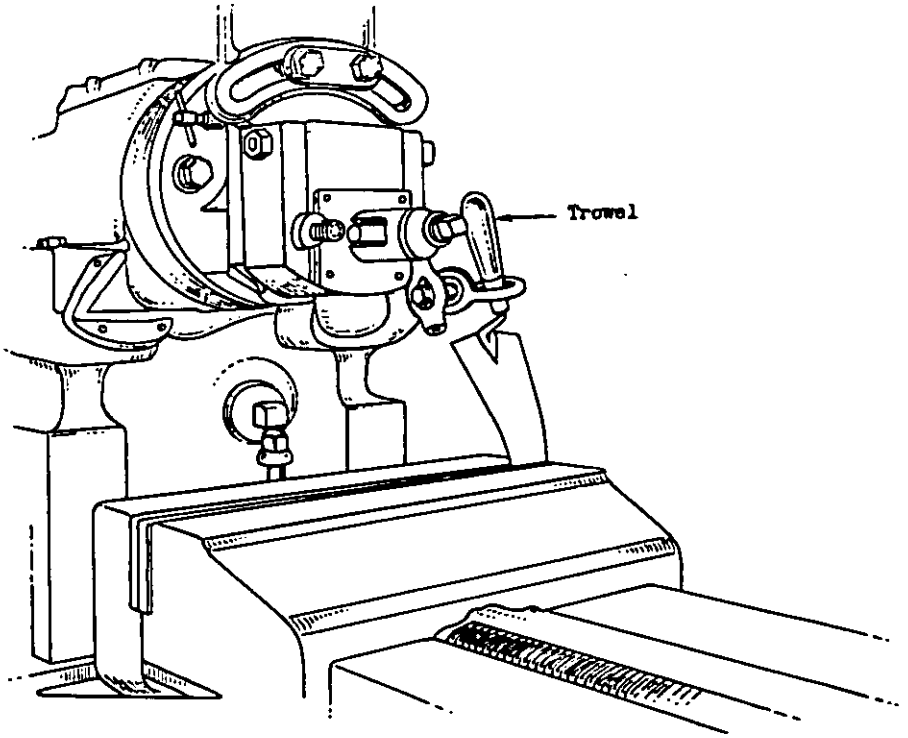


FIGURE 12.—Method of test for types III, IV, and VI through IX trowels, inclusive.

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4.4.2.2 Types III, VII, and IX trowels. The trowels shall be held in a vertical position with the point or toe 1-1/2 inches below the top surface of the vise jaws.

4.4.2.3 Types IV, VI, and VIII trowels. The trowels shall be held in a vertical position with the point or toe 1 inch below the top surface of the vise jaws.

4.4.2.4 The handles of types III, IV, and VI through IX trowels may be engaged by an eyebolt secured in a lathe dog which is secured to the tool post or by any similar device. The flexing test shall be as illustrated in figure 12 and conform to table I (see 3.6).

4.4.3 Type III trowel. Type III trowels shall be tested by cutting not less than six face bricks in half.

4.4.4 Plastic handle tests.

4.4.4.1 Flammability test. Handle shall be clamped in a vise and a flame from blowtorch, Bunsen burner or equivalent, shall be applied to one end for 30 seconds. If this does not ignite the material, flame shall be applied the second time for 30 seconds. If the second application does not ignite the handle, it shall be considered as having passed the test. If the material ignites and the rate of burn does not exceed 2 inches per minute it shall be considered slow burning, thereby also passing the test.

4.4.4.2 Shatter test. Handle shall be placed on a solid foundation and a suitable guard shall be placed around the handle. A weight of 5 pounds shall be dropped directly upon the handle from a 3-foot height. If the handle is still intact after the two direct blows, it shall have passed the test.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.1), the trowels shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing shall be A, B, or C as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6.2 Supersession data. Types and classes of trowels covered in this specification corresponding to those in GGG-T-671a are as follows:

GGG-T-671a	GGG-T-671b
Type I. - Plastering.	Type I. - Plastering.
Style A. - Malleable iron or steel mounting.	Deleted.
Style B. - Aluminum mounting.	Type I. - Plastering.
Type II. - Cement.	Type II. - Cement.
Class 1. - Midget.	Class 1. - Midget.
Style A. - Malleable iron or steel mounting.	Deleted.
Style B. - Aluminum mounting.	Type II, class 1.
Class 2. - 2. - Regular.	Class 2. - Regular.
Style A. - Malleable iron or steel mounting.	Deleted.
Style B. - Aluminum mounting.	Type II, class 2.
Type III. - Brick.	Type III. - Brick.
Type IV. - Pointer.	Type IV. - Pointer.
Type V. - Mastic (notched).	Type V. - Mastic (notched).
Class 1. - Linoleum.	Class 1. - Linoleum.
Style A. - Malleable iron or steel mounting.	Deleted.
Style B. - Aluminum mounting.	Type V, class 1.
Class 2. - Asphalt tile.	Class 2. - Asphalt tile.
Style A. - Malleable iron or steel mounting.	Deleted.
Style B. - Aluminum mounting.	Type V, class 1.
Type VI. - Gauging.	Type VI. - Gauging.
Type VII. - Pipe covering.	Type VII. - Pipe covering.
Type VIII. - Margin.	Type VIII. - Margin.
Type IX. - Tuck pointer, joint filler, or caulking.	Type IX. - Tuck pointer, joint filler, or caulking.

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